6-DOF Seed Dynamics: Physics of the Strip-Theory Model

This note describes the physics of the current 6-DOF (13-state) strip-theory seed model: what is carried over from the 2D Andersen–Pesavento–Wang/Pomerenk model (`minimal_imp`), what is genuinely new in 3D, and what has been added on top of both. It supersedes `seed3D_physics.tex`. Badge key: **[3D]** new in 3D but still traceable to the 2D physics; **[omitted]** a 2D term deliberately dropped (negligible in air, zero for the symmetric seed); **[heuristic]** a new heuristic term with *no* `minimal_imp` analogue and a weaker empirical basis (carries a tuning coefficient); **[toggle]** off by default, enabled explicitly.

Setup and conventions

The plate is sliced into spanwise *strips*; each strip is a local 2D cross-section using the exact 2D coefficient laws. Body axes: x = chordwise, y = plate-normal, z = spanwise. Everything is referenced to the center of mass (CoM). State: $[\,\mathbf{r}$ (CoM, inertial); $\mathbf{q}$ (body $\to$ world); $\mathbf{v}$ (inertial); $\boldsymbol{\omega}$ (body)$\,]$.

Two fixed body-frame references matter and must not be confused: the *geometric center* (strip mid-chord / centroid, fixed in the body) is where velocity is sampled and where the aerodynamic force magnitude/direction are set; the *center of pressure* (CoP, migrates with α) is used *only* as a moment arm. Per strip we use the in-plane relative velocity at the geometric center $\mathbf{v}_{ip} = (v_x, v_y)$ with speed $|v_{ip}|$, the local angle of attack $\alpha = \operatorname{atan2}(v_y, v_x)$, and the spanwise spin ω_z. Coefficients $C_T, C_D, l_{cp}, C_R, C_{D2}, C_{D0}$ come from `computeAeroCoeffs`. A set of dimensionless tuning knobs (default 1, no `minimal_imp` analogue) scale the optional / heuristic terms: C_{fy} (T_y damping), C_{Tx} (T_x damping), C_{span} (span force), $C_{\text{span},\tau}$ (span torque), and k_0 (span-torque roll-off).

Aerodynamic coefficients vs. angle of attack

These are a direct port of the 2D Andersen–Pesavento–Wang coefficient laws (`computeAeroCoeffs`); each strip evaluates them at its own α. Two tanh weights blend an *attached* model (small $|\alpha|$) into a *separated* model (large $|\alpha|$) around stall angle a_s with width d:

$$f_1 = \tfrac{1}{2}\Big[1 - \tanh\tfrac{|\alpha|-a_s}{d}\Big] \text{ (attached near } \alpha{=}0), \qquad f_2 = \tfrac{1}{2}\Big[1 - \tanh\tfrac{\pi-|\alpha|-a_s}{d}\Big] \text{ (mirror near } \alpha{=}\pm\pi).$$

Three branches in α then select the coefficients: $m_1 : |\alpha| \le \frac{\pi}{2}$ (attached / forward); $m_2 : \alpha > \frac{\pi}{2}$ (reversed, front); $m_3 : \alpha < -\frac{\pi}{2}$ (reversed, back). Drag and CoP are identical on m_2, m_3; only the lift sign differs.

Lift / normal-force C_T. Attached $\propto \sin\alpha$, separated $\propto \sin 2\alpha$:

$$\begin{aligned}
m_1 &: C_T = C_{L1} f_1 \sin\alpha + (1{-}f_1) C_{L2} \sin 2\alpha,\\
m_2 &: C_T = -\big[C_{L1} f_2 (\pi{-}|\alpha|) + (1{-}f_2) C_{L2} \sin 2(\pi{-}|\alpha|)\big],\\
m_3 &: C_T = +\big[C_{L1} f_2 (\pi{-}|\alpha|) + (1{-}f_2) C_{L2} \sin 2(\pi{-}|\alpha|)\big].
\end{aligned}$$

Drag C_D. Attached C_{D0} plus induced $\propto \sin^2\alpha$; separated $\propto \sin^2\alpha$:

$$\begin{aligned}
m_1 &: C_D = f_1\big(C_{D0} + C_{D1}\sin^2\alpha\big) + (1{-}f_1) C_{D2} \sin^2\alpha,\\
m_{2,3} &: C_D = f_2\big(C_{D0} + C_{D1}(\pi{-}|\alpha|)^2\big) + (1{-}f_2) C_{D2} \sin^2(\pi{-}\alpha).
\end{aligned}$$

Centre of pressure l_{cp} (fraction of chord from the geometric center; attached branch quadratic in angle, separated branch a linear ramp):

$$m_1 : l_{cp} = f_1\big(C_{CP0} - C_{CP1}\alpha^2\big) + (1-f_1)\big(C_{CP2} - \tfrac{C_{CP2}}{\pi/2}|\alpha|\big),$$
$$m_{2,3} : l_{cp} = -\Big[f_2\big(C_{CP0} - C_{CP1}(\pi-|\alpha|)^2\big) + (1-f_2)\big(C_{CP2} - \tfrac{C_{CP2}}{\pi/2}(\pi-|\alpha|)\big)\Big].$$

Constants (alpha-independent). C_R (rotational lift), $C_{D,\mathrm{rot}} = C_{D2}$ (spin damping), and C_{D0} (also used by the T_y heuristic). Default fit values (`minimal_imp`, CGS-model, dimensionless except a_s, d in rad): C_{L1}=5.2, C_{L2}=0.95, C_{D0}=0.1, C_{D1}=5, C_{D2}=1.9, C_R=1.1, C_{CP0}=0.3, C_{CP1}=3.5, C_{CP2}=0.2, a_s=14°, d=6°.

Preserved 2D quirks (kept identical so the 3D model reduces exactly to 2D): C_{T2}, C_{T3} use $(\pi-|\alpha|)$ where C_{T1} uses $\sin\alpha$ (a small-angle approximation of $\sin(\pi-|\alpha|)$ near $\alpha=\pm\pi$); C_{D2} mixes $(\pi-|\alpha|)^2$ with $\sin(\pi-\alpha)$. Do not "correct" these without a deliberate decision.

Per-strip forces

Each strip of chord c and width dz contributes the following, evaluated from the strip's geometric-center velocity. Lift is $\perp$ to the relative velocity, drag is $\parallel$ to $-\mathbf{v}$. Forces are summed over strips; gravity is added once at the end.

Force	Rough magnitude & direction	Acts at
Translational lift	$\frac{1}{2}\,\rho\,c\,dz\,C_T\,\lvert v_{ip}\rvert^2,\ \perp \mathbf{v}$	center of pressure, $x_{cp} = x_{gc} + l_{cp}\cdot c$ (migrates with α)
Translational drag	$\frac{1}{2}\,\rho\,c\,dz\,C_D\,\lvert v_{ip}\rvert^2,\ \parallel -\mathbf{v}$	center of pressure (same as lift)
Rotational lift	$\frac{1}{2}\,\rho\,c^2\,dz\,C_R\,\omega_z\,\lvert v_{ip}\rvert,\ \perp \mathbf{v}$	geometric center / mid-chord ($l_{cr} = 0$)

Translational lift — circulatory force from the flow turning around the plate; the workhorse that holds the seed up.
Translational drag — pressure/separation resistance opposing motion.
Rotational lift — Magnus-like lift from the plate's own spin shedding circulation. **[3D]** In 3D only the spanwise spin ω_z drives it (the in-plane rotation each strip feels). It *adds to the force* in the lift direction but its *torque uses a different arm* (mid-chord, not CoP), so `computeStripForces` returns it separately.

Per-strip torques (about the CoM)

A force at position $\mathbf{r}$ gives torque $(\mathbf{r} - \mathbf{c}) \times \mathbf{F}$ automatically—the cross product reproduces the 2D moment arm $l_\tau = l_{cp} - l_{cm}$ with no special handling.

Torque	Rough equation	Arm / axis
Lift+drag moment (T_t)	$(\mathbf{r}_{cp} - \mathbf{c}) \times \mathbf{F}_{\text{lift+drag}}$	arm = CoP − CoM
Rotational-lift moment (T_{cr})	$(\mathbf{r}_{gc} - \mathbf{c}) \times \mathbf{F}_{\text{rot lift}}$	arm = geometric center − CoM
Spin damping (T_r)	$-\frac{1}{128}\,\rho\, c^4\, dz\, C_{D2}\, \omega_z\lvert\omega_z\rvert \cdot \big[(\frac{2l_{cm}}{c}+1)^4+(\frac{2l_{cm}}{c}-1)^4\big]$	about spanwise (z) axis
Buoyancy couple (T_b) **[omitted]**	$-m_p\, g\, l_{cm} \cos\theta$	weight at CoM vs. buoyancy at mid-chord

T_t — pitching torque from lift+drag acting ahead of/behind the CoM; the dominant driver of flutter vs. tumble.
T_{cr} — moment of the rotational lift; kept separate because it acts at the geometric center ($l_{cr} = 0$), a different chordwise point than T_t.
T_r — nonlinear spin damping about the spanwise axis: each chordwise element sees drag $\propto (\omega_z r)^2$, and integrating $r\times$drag along the chord opposes spin. This is *separate physics*, not captured by the strip lift/drag (which are evaluated at the geometric center where ω_z contributes no velocity).
T_b **[omitted]** — pendulum-like restoring couple from weight and buoyancy at different points. Negligible in air ($m_p = \rho_{\text{air}} V \ll m$).

Whole-seed effects (evaluated once, not per strip)

The per-strip loop leaves two gaps: it only damps spin about the spanwise axis, and it produces *zero* spanwise (z) force. The strip lift/drag are sampled at each strip's geometric center, where spin about x or y contributes no in-plane velocity, so the earlier assumption that "damping about the other axes is captured implicitly by $\boldsymbol{\omega} \times \mathbf{r}$" is *false*—rotation about x and y was effectively undamped, and pure spanwise sliding felt no force. The following whole-seed terms fill those gaps. Each is a single contribution for the entire seed (see § "why once, not per strip" below), added after the strip loop.

Spin damping about the remaining axes

Torque	Rough equation	Notes
Chordwise spin damping (T_x) **[3D]**	$-\frac{1}{128}\,\rho\,S^4\,\bar{c}\,C_{D2}\,\omega_x\lvert\omega_x\rvert\cdot\big[(\frac{2l_{cm,z}}{S}+1)^4+(\frac{2l_{cm,z}}{S}-1)^4\big]$	exact T_r integral with the roles swapped: total span S plays the "chord", mean chord $\bar{c}$ plays dz, and the lever is the CoM's *spanwise* offset $l_{cm,z}$ from the span geometric center.
Normal spin damping (T_y) **[heuristic]**	$-\frac{1}{128}\,\rho\,R^4\,C_{D0}\,C_{fy}\,\omega_y\lvert\omega_y\rvert, \quad R=\frac{S}{2}$	No clean closed form: in-plane spin sweeps the plate edge-on (the ≈ 0 added-mass mode), so this is a placeholder that reuses the same $\frac{1}{128}\rho\,(\text{length})^4$ scaling and the measured C_{D0}, times a tuning knob C_{fy}. Meant to stay much smaller than T_x, T_r (the plate spins easily about its normal).

Rotation about the chordwise axis (ω_x) induces a velocity that varies with *spanwise* position, so its damping integral runs over the span—the direct analogue of T_r's chordwise integral, evaluated once over S. Rotation about the normal axis has no equivalent (the flow is edge-on both ways), hence the heuristic form.

T_x **is gated off by default** (`enableTxDamping`, scaled by C_{Tx}). The strips already capture roll damping through their spanwise-distributed normal loading—strips offset in span see $v_y = -\omega_x z$, go broadside, and their summed moment reproduces essentially the same span integral—so applying T_x on top roughly *double-counts* it. T_y has no such strip analogue and is always applied.

Spanwise-flow force [toggle] [heuristic]

Optional, experimental extension; the validated baseline model runs with the spanwise force off (it is toggled by `enableSpanForce`, with the further switches `enableSpanGeomVelocity`, `enableSpanCOPMigration`, `enableSpanTorqueAttenuation`). When enabled, a single whole-seed force models flow in the span–normal (z–y) plane, exactly the strip lift/drag law with the *span* S as the "chord" and the mean chord $\bar{c}$ as the width (so $S\bar{c}$ = wing area). The bulk velocity is transported to the seed geometric center $\mathbf{g}$,

$$\mathbf{v}_{gc} = R^\top \mathbf{v} + \boldsymbol{\omega} \times (\mathbf{g} - \mathbf{c}), \qquad \beta = \operatorname{atan2}(v_{gc,y},\, v_{gc,z}), \quad |v| = \sqrt{v_{gc,z}^2 + v_{gc,y}^2},$$

and with $L_0 = \frac{1}{2}\rho S\bar{c}\, C_T(\beta)\, |v|\, C_{\text{span}}$, $D_0 = -\frac{1}{2}\rho S\bar{c}\, C_D(\beta)\, |v|\, C_{\text{span}}$ the full force in the body frame is

$$\mathbf{F}_{\text{span}}^{\text{full}} = \big[\, 0,\ \underbrace{-L_0 v_{gc,z} + D_0 v_{gc,y}}_{\text{normal }(y)},\ \underbrace{L_0 v_{gc,y} + D_0 v_{gc,z}}_{\text{span }(z)} \,\big]^\top.$$

Force: keep only the spanwise component. The body-z direction was previously unmodeled, so injecting $F_{\text{span},z}$ cannot double-count anything. The body-y (normal) component is *discarded from the force sum*: the strips already supply the normal force, and re-adding it here would double-count. (Body-x is identically zero.) A purely in-plane trajectory has $v_{gc,z} = 0 \Rightarrow \beta = \pm 90^\circ \Rightarrow C_T \approx 0 \Rightarrow F_{\text{span},z} \approx 0$, so this force vanishes for in-plane motion and does not perturb the 2D-reducible behavior.

Offset spanwise torque: use the full force. The torque *does* use the full force (including the discarded normal component) against a *migrating span-CoP* $z_{cp} = z_{gc} + l_{cp}(\beta)\, S$:

$$\boldsymbol{\tau}_{\text{span}} = (\mathbf{r}_{cp} - \mathbf{c}) \times \mathbf{F}_{\text{span}}^{\text{full}}, \qquad \mathbf{r}_{cp} = \big[\, x_{gc,\text{chord}},\ 0,\ z_{cp} \,\big]^\top.$$

This is a deliberate modeling choice. The strips supply the normal force and its chordwise-CoP moment, but strip theory assumes uniform spanwise loading and can *never* produce the roll moment (about x) from a span-offset pressure center; crossing the span-CoP arm with the full force recovers exactly that missing roll moment, $\tau_x = -r_{cp,z}\, F_{\text{span},y}$. Note that with only the z-force this term would be inert—the span-CoP migrates along z, *parallel* to that force, so its arm would contribute nothing and $\boldsymbol{\tau}_{\text{span}}$ would collapse to a pure yaw from any chordwise CoM offset. The application point is switchable (`enableSpanCOPMigration`: migrating span CoP, or the fixed geometric center), and the torque carries an *independent* scale $C_{\text{span},\tau}$ so force and moment tune separately: $\boldsymbol{\tau}_{\text{span}} \to C_{\text{span},\tau}\, a(k)\, \boldsymbol{\tau}_{\text{span}}$.

Reduced-frequency attenuation of the span torque. The span torque is a lumped *quasi-steady* CoP moment, valid only when the span-plane flow direction changes slowly relative to the convective time $S/|v|$. In autorotation the descent velocity resolved into the spinning body frame makes β—and hence this torque—oscillate at the normal-axis spin ω_y, parametrically destabilising roll. It is therefore scaled by a reduced-frequency factor

$$a(k) = \frac{1}{1 + (k/k_0)^2}, \qquad k = \frac{|\omega_y|\, S}{2\,|v|},$$

keyed on ω_y (the autorotation spin) rather than $|\boldsymbol{\omega}|$: this leaves the tumbling mode (an ω_z mode, small ω_y) unattenuated while suppressing the oscillating torque during fast autorotation. Toggled by `enableSpanTorqueAttenuation`; $k_0 \approx 0.2$ preserves tumbling.

Why once, not per strip. T_r is summed per strip because its chordwise integral is identical at every span station, so dz-weighting reconstructs the full double integral. T_x and the span force are the opposite: span *is* the axis the strip loop discretizes, so re-integrating over the span per strip would replay (and badly overcount) the whole-span integral. They are therefore computed once, using seed-wide aggregates (S, $\bar{c}$, wing area, and the area-weighted geometric center).

Added mass

Fluid the plate must accelerate with it; scales with fluid density, so small in air but retained. Built per strip and summed (`getAddedMass`).

Block	Content
Translational A_{trans}	$\text{diag}\big(0,\ \sum \pi\rho(c/2)^2 dz,\ 0\big)$: only broadside (normal, y) entrains fluid; edge-on (x) and in-plane span (z) ≈ 0.
Rotational A_{rot}	$I_{zz} = \sum(i_a + m_2 x_s^2)$ (the 2D I_a); $I_{xx} = \sum m_2 z_s^2$ **[3D]**; $I_{yy} \approx 0$ **[3D]**; $I_{xz} = -\sum m_2 x_s z_s$ (zero if symmetric).

Here $m_2 = \pi\rho(c/2)^2 dz$, $i_a = \pi\rho c^4 dz/128$, and x_s, z_s are the strip's chordwise/spanwise offsets from the CoM.

What changes between 2D and 3D

Reference frame / Coriolis terms. The 2D code works in the body frame, so it carries rotating-frame terms $(m{+}m_2)\omega v_{yp}$, $-(m{+}m_1)\omega v_{xp}$. We sum translational forces in the *inertial* frame, where these vanish identically—provided A_{trans} is rotated into the inertial frame, $R\,A_{\text{trans}} R^\top$, before dividing.

Added-mass CoM-offset coupling. **[omitted]** The 2D terms $-m_2\omega^2 l_{cm}$ and $+m_2\dot{\omega}\, l_{cm}$ (translation↔rotation added-mass coupling) are dropped. Zero for the symmetric seed ($l_{cm} = 0$) and $\propto \rho_{\text{air}}$ otherwise.

Gyroscopic torque. **[3D]** The rotational equation gains $\boldsymbol{\omega} \times (I\boldsymbol{\omega})$, identically zero for scalar 2D rotation but real in 3D (spinning about one axis induces torque about others).

Time-varying inertia $\dot{I}_G$. **[3D]** If the CoM drifts (e.g. a shifting nut), I_G changes in time, adding $\dot{I}_G\,\boldsymbol{\omega}$ to the rotational equation. 2D inertia was constant.

Velocity sampling point. 2D evaluates the reference velocity at the CoM ($v_{yp} - \omega l_{cm}$); per strip we evaluate at the strip *geometric center* and let $\boldsymbol{\omega} \times \mathbf{r}$ supply the rotational part. This is why T_r (and now T_x) must stay explicit—geometric-center sampling misses the spin-drag distribution.

Attitude. [3D] A full quaternion replaces the single angle θ; orientation advances by $\dot{\mathbf{q}} = \frac{1}{2}\,\mathbf{q} \otimes [0, \boldsymbol{\omega}]$.

Effective weight. 2D uses $(m - m_p)g$ (weight minus buoyancy); we use $m\,g$, consistent with dropping buoyancy.

Whole-seed damping and spanwise force. [3D] [heuristic] [toggle] New relative to both models: explicit damping about x and y (T_x, T_y) and an optional spanwise-flow force with an offset roll torque, all described above. T_x is traceable to T_r; T_y and the span force are heuristics with tuning knobs.

Reproducing the flight modes

The 2D-reducible baseline (spanwise force off) captures gliding, diving, and fluttering directly. Reproducing the full biological set of 3D modes—including the large tumbling spirals—uses the whole-seed spanwise extensions with the empirically-tuned configuration below (recorded in `testing/Working_Dynamics_`
All runs use the base seed (nut mass $75\,\mu$g at the body center, body density $65\,\mathrm{kg/m^3}$, span $S = 50\,\mathrm{mm}$, chord $c = 15\,\mathrm{mm}$) in air ($\rho = 1.225$, buoyancy ignored), released from rest for $10\,\mathrm{s}$ unless an initial attitude is given.

Physics switches. `enableSpanForce`, `enableSpanGeomVelocity`, `enableSpanCOPMigration`, and `enableTxDamping` all *on*; `enableSpanTorqueAttenuation` *off*. Notably the reduced-frequency attenuation is *not* used in this tuning: stability is instead obtained with a weak span force ($C_{\mathrm{span}} = 0.2$) plus roll damping (T_x) on, which raises the parametric-instability threshold directly.

Tuned coefficients (all others at the `minimal_imp` defaults): $C_{\mathrm{span}} = 0.2$, $C_{\mathrm{span},\tau} = 0.7$, $C_{fy} = 1$, $C_{Tx} = 1$, $k_0 = 0.2$.

Mode	**Nut position** $[x; y; z]$	**Initial condition**
Spanwise-axis fluttering	$[0;\ 0;\ 0]$ (center)	$\frac{\pi}{6}$ tilt about z
Gliding	$[0.5c;\ 0;\ 0]$	from rest
Diving	$[1.5c;\ 0;\ 0]$	from rest
Fluttering + spiral	$[0;\ 0;\ 0.01S]$	$\frac{\pi}{6}$ tilt about z
Fluttering + tight spiral	$[0;\ 0;\ S]$	$\frac{\pi}{6}$ tilt about z
Autorotation	$[c;\ 0;\ 1.2S]$	from rest
Parachute	$[0;\ -c;\ 0]$	from rest

These are basic illustrative examples—not the exact mode-onset boundaries.

Where it fits in the implementation

Functions live under `physics/` with sub-folders `aero/`, `mass/`, and `helpers/`.

Function	Role
`seed6DOFODE`	main ODE right-hand side; orchestrates everything below
`setupSeedShapeAndMass`	builds strip geometry, CoM(t), $I_G(t)$, $\dot{I}_G(t)$ from a polyshape + nut
`translationDynamics` / `rotationDynamics`	solve $\mathbf{a}$ (inertial) and $\boldsymbol{\alpha}$ (body, modified Euler)
`mass/getMassProperties`	CoM $\mathbf{c}$, I_G, $\dot{I}_G$, added mass at time t
`mass/getAddedMass`	A_{trans}, A_{rot} (tables above)
`aero/computeAeroCoeffs`	$C_T, C_D, l_{cp}, C_R, C_{D2}, C_{D0}$ vs. α; tuning C_{fy}, C_{span}
`aero/computeAngleOfAttack`	$\alpha = \text{atan2}(v_y, v_x)$
`aero/computeStripCoP`	chordwise CoP $x_{cp} = x_{gc} + l_{cp}\, c$
`aero/computeStripForces`	per-strip lift+drag (T_t arm) and rotational lift (T_{cr} arm), returned separately
`aero/stripSpinDamping`	per-strip T_r (spanwise-axis spin damping)
`aero/spanSpinDamping`	whole-seed T_x (chordwise-axis spin damping) **[3D]**
`aero/normalSpinDamping`	whole-seed T_y (normal-axis spin damping) **[heuristic]**
`aero/computeSpanForce`	whole-seed spanwise-flow force + span-CoP fraction **[heuristic]** **[toggle]**
`helpers/computeSeedLocalVel`	per-strip body-frame velocity via $\mathbf{v} + \boldsymbol{\omega} \times \mathbf{r}$
`helpers/quatKinematics`	$\dot{\mathbf{q}}$ update
`helpers/quatToRotm`, `quatMultiply`	quaternion utilities